
Dashboard Review for the DOT Planner — Data Accuracy & Design Options

Working analysis and decision document. Citi Bike Rebalancing Explorer. Prepared July 2026. Nothing here is implemented yet — these are proposals to react to.

Summary

This document answers four things asked of the dashboard, in the order they should be read:

1. **How accurate the data and predictions actually are**, per Investigator feature — the grounded part, which should drive every design decision below.
2. **What the right sidebar should show** to be useful to a DOT planner.
3. **What value Investigator Mode can bring** to a planner checking scenarios.
4. **How to redesign the top-left controls** (Individual/Grouped + the time/period selector).

The headline finding, because it shapes everything else:

The dashboard is a *climatology* tool — what a station *typically* does — not a next-day forecaster.
Those are two different accuracy questions, and conflating them is the main way a planner could be misled by the numbers.

Part 1 — How accurate is the data, per feature

1.1 Single-day flow is almost pure noise; the *averaged* curve is what's reliable

At the raw (station, date, hour) grain, the mean absolute net flow is **2.23 bikes**, and the best forecast (seasonal-naive) has **MAE 1.95** — the error is roughly **88% of the signal**. No model beats it out-of-sample (that is the honest headline shown in the model panel). **You cannot tell a planner what a specific station will do next Tuesday at 8am.**

But the dashboard does not show single-day predictions — it shows the **averaged curve** (the trailing mean over many real days). How reliable that displayed number is depends entirely on the slice the planner has selected:

View (period selector)	Days averaged (median)	Std. error of the shown number	Stations whose deficit/surplus sign is statistically confident ($ \text{mean} > 2 \cdot \text{SE}$)
All-period average (default)	~84	0.19 bikes/hr	93%
Season	~22	0.34 bikes/hr	82%
Single month	~8	0.50 bikes/hr	60%

Implication: the default view is trustworthy for prioritization, but drilling down to a single month means ~40% of “signals” are no longer sign-confident. That is a concrete design cue (see Part 2C and Part 4).

1.2 The three Investigator features have fundamentally different accuracy *characters*

Feature	What it really is	How much to trust it
Equity thresholds	Deterministic geometry (haversine distances)	Exact counts. The uncertainty is <i>data</i> , not statistics: schools are a 2019–2020 vintage layer; 216 NYCHA developments / 1,899 schools / 2,120 subway entrances; and “within 300 m” ≠ “actually served.” Trust the number; caveat the data currency.
Fleet simulator	Deterministic <i>capacity math</i> on a reliable flag	The flag (which stations are AM-deficit) rests on the reliable all-period average, so <i>who needs a truck</i> is solid. But the serviced counts (14 → 145) come from a greedy heuristic with a placeholder depot and a 45-stop cap — read them as “ <i>how many stations N trucks × 45 stops can physically touch,</i> ” not an optimized operational plan. This is also why there are no diminishing returns.
Weather scenario	A genuine statistical model — precise but weak	The subtle one. Coefficients are <i>extremely</i> tight (temp +0.034/°C , CI $\approx \pm 0.0005$) because n is huge — but the regression’s R² is only 0.04–0.07 . Weather explains ~5% of day-to-day variance . So the scenario is a good estimate of the <i>average direction and rough size</i> (ideal→hot $\approx$ +11% magnitude; ideal→rain $\approx$ –15%), but a poor predictor of any single day. A tight confidence interval ≠ an accurate forecast — a planner will misread this unless it is shown as a directional band, not a precise number.

Trust summary for a planner: - **Equity** — trust the number (caveat the vintage). - **Fleet** — trust *who* needs service; treat *how many get serviced* as capacity illustration. - **Weather** — trust the *direction and rough size*, not the decimals.

1.3 Weather regression, in numbers

Typology group	n (station-days)	Temp coef (/°C)	Precip coef (/mm)	R²
Commuter core	230,915	+0.034	−0.017	0.059
Residential feeder	385,102	+0.037	−0.018	0.073
Pooled	616,017	+0.028	−0.014	0.044

Projected average magnitude for the presets (pooled fit): **Ideal (22 °C) $\approx$ 2.21 · Hot (31 °C) $\approx$ 2.45 (+11%) · Steady rain (15.6 °C, 10.2 mm) $\approx$ 1.88 (−15%)**. Real, directionally sensible, modest — and dwarfed by the ~95% of variance the model does not explain.

Part 2 — Right sidebar: what actually helps a DOT planner

Today the rail shows four readout tiles, two “worst now” lists, Investigator Mode, and the model panel. It is good, but it is missing the metrics a planner makes decisions on.

2A. Upgrade the readout tiles to be decision-relevant

The biggest gap: **imbalance and equity never meet**. The most valuable single number for this project’s own framing is the **intersection**:

- **Deficit stations that are *also* equity-priority** — e.g. “*of 1,037 AM-deficit stations, N are within 300 m of NYCHA / a school.*” This is the barrier-to-access number. Right now equity and deficit sit in separate tiles and never cross.
- **Total system pressure** — the sum of |deficit| bikes across flagged stations: one “how big is the problem” number that moves as the planner scrubs time.
- **Concentration** — “*top 20 stations = X% of total deficit,*” which tells a planner whether targeting a handful of stations is worthwhile.
- **Confidence / coverage** — “*this slice averages ~8 days*” when they drill to a month, so they know the number just got noisier.

2B. Make the “worst now” lists planner-grade

Add, per row: an **equity flag icon** (near NYCHA / school), and a **reliability marker** (is this a chronic imbalance or a noisy single-month reading?). Optionally a **peak hour** so they know *when* to send a truck.

2C. Add a confidence surface

When the sign is not statistically confident ($|\text{mean}| < 2 \cdot \text{SE}$, common at the month grain), de-emphasize or mark those stations rather than showing them as equal-weight deficits.

Part 3 — Investigator Mode: where the planning value is

Reframe the three existing controls around planner decisions, and consider new modes.

- **Fleet** → “**Coverage & gap**” tool. The most useful reframing: “*N trucks reach X of 1,037 deficit stations — and Y of the equity-priority ones. Here are the stations left unserved.*” The **residual gap** and the **equity reach** are the real planning outputs, not the serviced count alone.
 - **Weather** → “**Stress test.**” Given the low R^2 , do not sell precision. Sell “*on a hot day, how many more stations tip into deficit?*” as a **range**. Value = resilience planning, not point prediction.
 - **Equity thresholds** → “**Policy sensitivity.**” Value = “*if we define ‘near NYCHA’ as 500 m instead of 300 m, the flagged population changes by N*” — genuinely useful for setting the policy threshold itself.
 - **New ideas worth considering:** a **borough / district filter** (planners work by district); an “**expansion targeting**” mode (high transit-gap + equity + latent demand); a **peak-hour finder** (which hour has the worst system-wide imbalance).
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Part 4 — Top-left controls: redesign options

Problems today: Individual/Grouped is now *auto-by-zoom*, so the manual toggle is largely redundant; the period dropdown crams three granularities (All / seasons / months) into one flat list; and the temporal controls are split between the top-left card and the bottom time slider.

How comparable tools handle this (reference): transportation-planning tools like **Replica / StreetLight / Remix** separate “*what am I looking at*” (layer/view) from “*what time context*” (a dedicated date/time panel); weather scrubbers like **Windy** use a single timeline filmstrip; **MTA / Citi Bike** data views lean on a granularity segmented control (Year → Month → Day). The common pattern is **one coherent time-context panel, separate from view/layer controls**.

Option	Idea	Trade-off
A — Unified time-context panel	Group <i>all</i> temporal controls together: granularity as a segmented control (All · Season · Month) + a secondary picker + day-type + the hour scrubber, in one panel. View/route controls move elsewhere.	Cleanest mental model; biggest layout change.
B — Demote the view toggle	Since view is auto-by-zoom, replace the manual toggle with a small map chip (“Grouped — zoom in for detail”) that only reveals the override on click. Frees the card for temporal controls.	Small change, quick win; keeps most of the current layout.
C — Month filmstrip	Replace the period dropdown with a horizontal season/month strip you click across (and can animate), instead of a hidden dropdown.	Great for “watch the year change”; needs horizontal space.

Recommendation: B + a lighter version of A (demote the view toggle, and turn the period dropdown into a two-level segmented control) is the highest value-to-effort, paired with a **confidence indicator** driven by the reliability numbers in Part 1.

Decisions on the table

1. **Sidebar** — add the **imbalance × equity intersection** metric (top recommendation)? And reliability markers on the worst-now lists?
2. **Investigator** — reframe the existing three (coverage-gap / stress-test / policy-sensitivity), or also add a new mode (district filter / expansion targeting)?
3. **Weather honesty** — present it as a **directional band** rather than a precise % (given $R^2 \approx 0.05$)?
4. **Top-left** — Option A, B, C, or the B+A blend?

Appendix — how the accuracy numbers were computed (reproducible)

All figures derive from committed data; nothing is estimated.

- **Signal magnitude & MAE grain:** `data/daily_net_flow.parquet` (11,777,290 rows). Mean `|net|` and quantiles computed directly on the `net` column. The model MAE (naive 1.95) is scored in

`pipeline/demand_model.py` against the same single-day `net`, confirming the forecast-error-vs-signal ratio.

- **Displayed-curve reliability:** grouped `net` by `(station, [period], day_type, hour)`; standard error = $\text{std} / \sqrt{n_days}$; “sign-confident” = `|mean| > 2·SE`, restricted to actionable buckets (`|mean| ≥ 1`). Computed at three grains: all-period, season (quarter proxy), month.
- **Weather R²:** OLS of `magnitude ~ 1 + temp_mean_c + precip_mm` on `pipeline.elasticities.load_daily_weather_panel(...)` (788,979 station-days across 18 weather zones), per typology group and pooled; $R^2 = 1 - \text{SS_res} / \text{SS_tot}$.
- **Equity / fleet counts:** `data/flows.json` equity-join block, `data/route.json`, `data/fleet_scenarios.json`.

Reproduce the whole pipeline with `python3 -m pipeline.reproduce_all --skip-download`.